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Data Science FINAL PROJECT REPORT

Project Title:

Automated Multi-Class Classification of Brain Tumors Using Deep Learning and Explainable AI

Student Name and SRN: Jashwanth Reddy Gandavarapu - 24048724

Supervisor: Krishna kedia

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DECLARATION STATEMENT

This report is submitted in partial fulfilment of the requirement for the degree of Master of Science **in Data Science** at the University of Hertfordshire.

I have read the detailed guidance to students on academic integrity, misconduct and plagiarism information at Assessment Offences and Academic Misconduct and understand the University process of dealing with suspected cases of academic misconduct and the possible penalties, which could include failing the project or course.

I certify that the work submitted is my own and that any material derived or quoted from published or unpublished work of other persons has been duly acknowledged. (Ref. UPR AS/C/6.1, section 7 and UPR AS/C/5, section 3.6)

I did not use human participants in my MSc Project.

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Student Name printed: Jashwanth Reddy Gandavarapu

Student Name signature :

Student SRN number: 24048724

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Abstract

The diagnosis of brain tumors remains one of the most daunting challenges in oncology since early detection precision determines patient survival rates. I created an automated system for this research which uses deep learning and Explainable AI (XAI) to conduct brain tumor type classifications. Using pre-trained CNN architectures I applied VGG16, DenseNet121, and InceptionV3 models to detect Glioma, Meningioma, Pituitary, and No Tumor classes from MRI images. Through extensive data augmentation and fine-tuning I achieved my highest performance with InceptionV3 which reached an 85.13% accuracy rate. I used Grad-CAM to produce visual explanations which demonstrate tumor locations because I needed to achieve clinical transparency. My findings demonstrate that this system provides a dependable diagnostic aid which empowers medical professionals with interpretable high-accuracy insights for neurological care.

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Chapter 1: Introduction

Introduction

The MRI images of brain tumour classification are crucial processes in medical diagnostics, where the early and correct identification are important in enhancing the treatment outcome. Convolutional Neural Networks (CNNs) and other deep learning models have proved to be very capable in automating this process and minimizing diagnostic errors (Chen et al., 2025)

Background of Study

The recent progress in artificial intelligence has allowed creating automated brain tumour segmentation and classification systems. MRI images can be trained using deep learning methods such as CNNs and hybrid models to identify a wide range of types of tumours. Attention mechanism studies combined with transformer-based models have also enhanced feature extraction and classification accuracy to reach a performance with more than 96% in multi-class classification tasks (Zarenia et al., 2025)

Present Market Practice

In the modern clinical setting, brain tumour diagnosis is mostly dependent on the manual or computer-aided diagnosis (CAD) systems of radiologists using MRI scans. Though CAD tools can help in enhancing efficiency, they tend to be weak and unclear. The current trends in the industry indicate that more diagnostic tools based on deep learning are being adopted, and most of these systems are still configured as a black-box system, which restricts trust and understandability of clinical decisions (Gomez-Guzman et al., 2025).

Problem Statement

Although the deep learning models have a high accuracy rate, there are still a number of challenges in automated brain tumour classification. The available models are usually faced with the challenge of class imbalance, lack of diversity of the dataset and interpretability. Additionally, most of them are concerned with classification only, as opposed to visual descriptions of predictions, which is essential to clinical validation. Lack of the Explainable AI (XAI) methods makes people less confident in automated systems, and it is hard to make the health professionals trust such models in practice ((Wong et al., 2025) et al., 2024).

Aim and Purpose

This proposed study is expected to create a multi-class brain tumour classifier based on deep learning models with Explainable AI tools to enhance accuracy, interpretability, and clinical reliability.

Objectives

- To predict and compare various pre-trained CNN models (e.g., VGG16, ResNet50, EfficientNet).
- To use progressive data augmentation methods to enhance model performance.
- To assess model in terms of accuracy, precision, recall, and F1-score.
- To add Explainable AI methods (e.g., Grad-CAM) to interpret the models.
- To examine multi-class brain tumour dataset model performance.

Research Questions

1. What pre-trained Convolutional Neural Network (CNN) model (e.g., VGG16, ResNet50, or EfficientNet) can classify best in distinguishing between Glioma, Meningioma, Pituitary and No Tumor classes in terms of classification accuracy and F1-score?
2. Would the sensitivity of the model (recall) in the under-represented tumor classes greatly increase with the use of more sophisticated data augmentation methods than with the base model trained on the raw dataset?
3. Is Explainable AI (XAI) technique, which is Grad-CAM, able to correctly locate the tumor region in the MRI scan, and do they make visualizations that are expected by radiologists?

Structure of Report

This report is divided into several chapters. Chapter 1 presents the topic of the research and its objectives. The literature review is presented in Chapter 2. Chapter 3 outlines the dataset and processing of data. The Ethical Consideration is described in chapter 4. Chapter 5 talks of the details of implementation. Chapter 6 includes results. Chapter 7 is a discussion and analysis, and Chapter 8 is the conclusion of the work and recommendations.

Chapter 2: Background

Introduction

The Deep Learning (DL) application in medical imaging has transformed the diagnostic process of the neurological pathology. This is a literature review that assesses the current computational methodologies of brain tumor multi-classification (based on Magnetic Resonance Imaging or MRI). The papers were selected according to three main requirements: the articles had to be devoted to the issue of the four-class classification (Glioma, Meningioma, Pituitary and No Tumor), to the use of various Convolutional Neural Network (CNN) architectures, and to be published in peer-reviewed journals in 2024-2025. Such studies are the best in the Data Science field of transfer learning and object detection.

Critical Review of the Major Research.

Study 01

Iqbal et al. (2024) have explored the application of the EfficientNet-B0 architecture, which is optimized to be used in the Internet of Medical Things (IoMT). The study used MRI datasets to identify four stages of tumors and the study was remarkable in its accuracy at 99%.

Critical Analysis: This work is strong because it balances the high performance and computing efficiency to be used in the edge-computing in healthcare. Nevertheless, the Pituitary and No-Tumor classes were found to have high F1-scores (0.96 and 0.94 respectively), but the paper is somewhat limited as to not describing the specific hardware limitations of the IoMT environment it alleges to be running on. This is relevant to my project as it shows that lightweight models can be as good as deeper architecture in terms of performance.

Study 02

Disci et al. (2025) has done a detailed comparative analysis of various pre-trained models such as Xception, MobileNetV2, and ResNet50 on a dataset of 7,023 images. Their results found the best model to be Xception whose weighted accuracy is 98.73.

Critical Analysis: The paper is scholarly as it benchmarks various architectures. One of the contributions includes their clear evaluation of the shortfalls of the black-boxes and the general challenge of enhancing the recollection of Glioma and Meningioma. One of the constraints is that it has not been clinically validated but in this project, it gives a crucial point of reference in selecting Xception as an effective backbone in feature extraction.

Study 03

Moving away as far as pure classification to localization, Nimmagadda & Devi (2025) ran YOLOv7 to a Roboflow dataset of 2,870 images. They attained 0.879 mAP (mean Average Precision) at the 0.5 IoU.

Critical Analysis: Compared to the other articles, the article is dedicated to the detection of bounding-box which plays a significant role in the localization of tumors. Although the box detection accuracy (0.837) is not as high as pure classification accuracies in other domains, it also provides additional spatial data to radiologists. The weakness is that the mAP (0.442) decreases drastically at larger IoU levels indicating that the model has difficulty in defining boundaries precisely- a technical fault my project will look at taking into account.

Study 04

Wong et al. (2025) used the base model of VGG16 but focused on expanding the dataset, through the use of augmentation to achieve 17,136 images. This method presented the best accuracy of 99.24.

Critical Analysis: This paper demonstrates that data volume and diversity are usually more important than architectural complexity. They also applied a web-based deployment with Dash which gives a template of the end-user interface. The weakness is the use of VGG16 which is more computationally costly to train compared to EfficientNet or MobileNet that have been mentioned earlier.

Summary of Findings

The literature shows that there is an agreement that transfer learning is the gold standard in MRI analysis. Accuracies are also very high, but there is a repetitive technical problem; the difference in recall between types of tumors and the requirement of model interpretability. Table 2.1 shows the summarized Literature review

Table 2.1: Summarized Literature review

Study	**Primary Model**	**Key Metric**	**Focus**
Iqbal et al. (2024)	EfficientNet-B0	99% Accuracy	IoMT/Efficiency
Disci et al. (2025)	Xception	98.73% Acc	Model Comparison
Nimmagadda (2025)	YOLOv7	0.879 mAP	Localization
Wong et al. (2025)	VGG16	99.24% Acc	Data Augmentation

Chapter 3: Dataset

Dataset Overview

The Brain Tumor MRI[1] Dataset is used which is available to the public for their studies on multi-class brain tumour classification. The dataset consists of MRI brain images collected from clinical environments and research institutions. Medical professionals annotated the images which were captured during neurological examinations with MRI scanners. The data was compiled to support research in automated diagnosis and improve tumour detection accuracy.

The dataset includes four classes: Glioma, Meningioma, Pituitary, and No Tumour.

Dataset Description

The dataset contains labeled MRI images organized into class-wise folders. Each image represents a 2D brain scan in standard image formats.

Key characteristics:

- The study includes four tumor classifications.
- The research uses labeled images to teach artificial intelligence through supervised learning.
- The study uses class distribution which shows balanced proportions between different groups.
- The dataset works well with CNN-based classification methods and XAI analysis techniques

Justification for Dataset Selection

This dataset was chosen because it directly supports multi-class classification tasks required for this study. The system provides sufficient data to train deep learning models which contain clinically important tumor types. Its structure also supports Explainable AI techniques such as Grad-CAM.

Data Loading and Preprocessing

I uploaded and extracted the dataset into training and testing directories. I created a special function that uses Keras functions to load MRI images from the system. The system converted each image into numerical arrays after resizing it to 128×128 pixel dimensions. I achieved pixel value normalization through division by 255 which resulted in values that ranged between 0 and 1, thus enhancing model training efficiency.

[1] https://data.mendeley.com/datasets/zp67tkpj2y/1

The system assigned numerical class labels according to folder names and converted these labels into one-hot encoded vectors through to_categorical() function for multi-class classification purposes.

Exploratory Data Analysis

Exploratory Data Analysis (EDA) is the process of examining and summarizing a dataset using statistics and visualizations to understand its structure, patterns, and quality before applying machine learning models.

Sample Dataset

Figure 3.1 shows the sample dataset i.e. which doesn't contain tumor in the brain

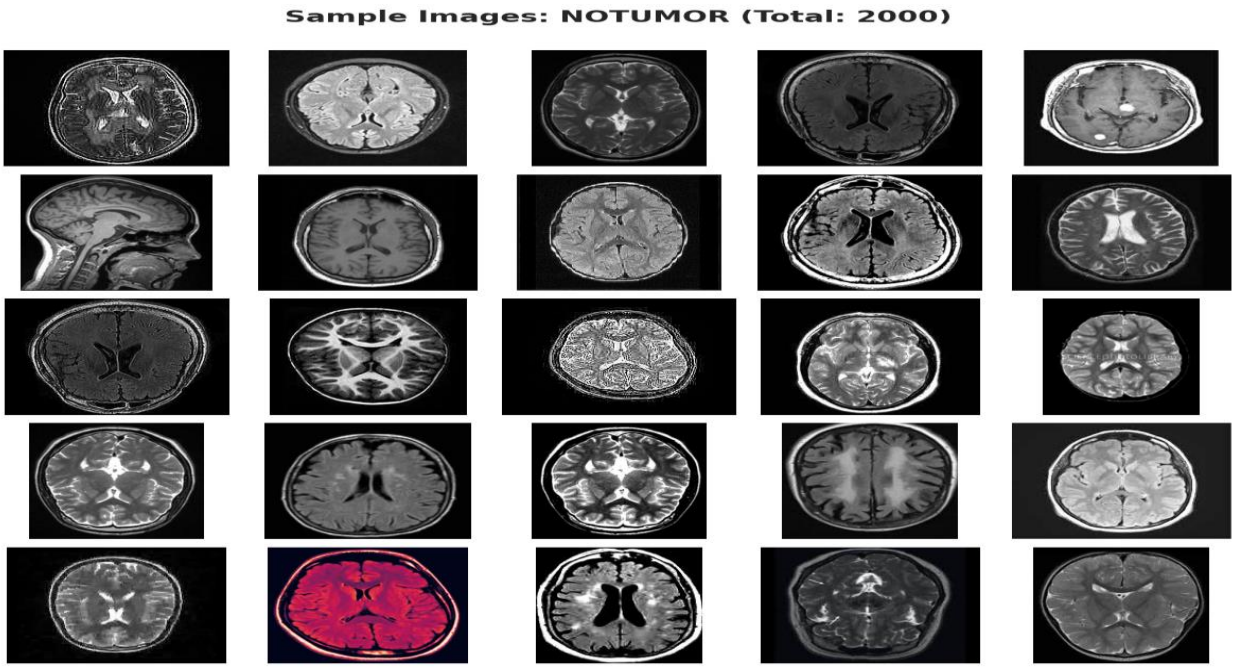

Figure 3.1: Sample image from dataset

Class Distribution

The dataset in Figure 3.2 exhibits nearly equal distribution between its multiple classes. The No Tumor class has approximately 2000 images which represent its highest count while Pituitary has about 1750 images and Meningioma has about 1650 images and Glioma has about 1600 images which represent their lowest counts.

I used data augmentation to create better class representation because the slight class imbalance would decrease model performance.

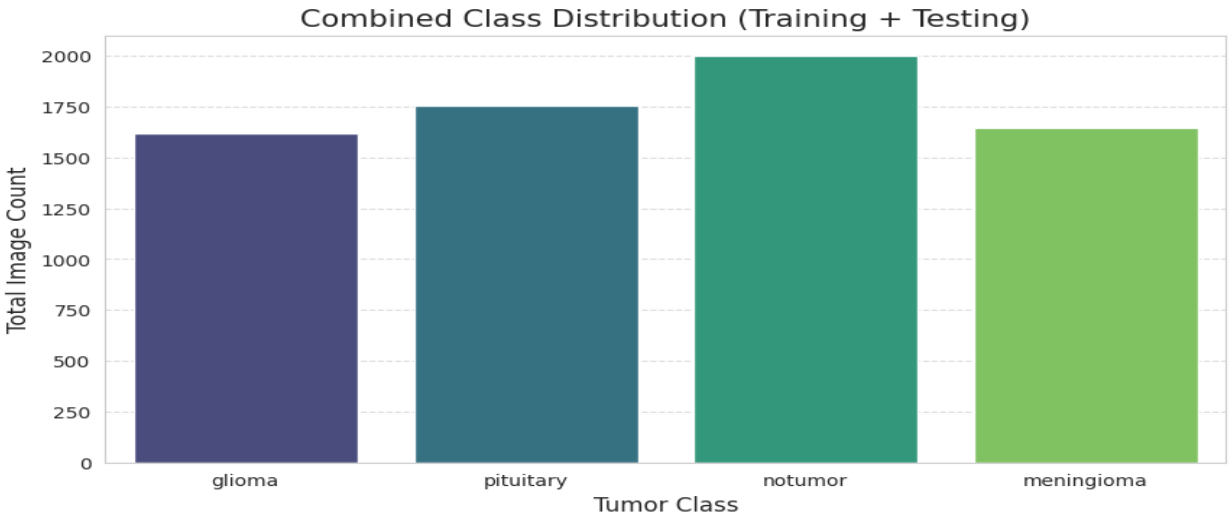

Figure 3.2: Class Distribution Graph

Chapter 4: Ethical Considerations

Personal Data and Privacy

The Brain Tumour MRI dataset contains only anonymized medical images with no personally identifiable information. The dataset protects patient confidentiality because it does not contain any information which would create privacy concerns.

GDPR and Ethical Approval

The dataset does not require GDPR compliance because it contains no personal data. The project does not need ethical approval because it uses publicly accessible secondary data without any human participation or survey research or data collection activities.

Data Licensing and Permission

The dataset is obtained from Mendeley Data under a CC BY 4.0 license which permits free distribution and modification of the content when users provide appropriate credit. The statement confirms that the content has been used according to legal and ethical standards without needing to make any payments.

Ethical Data Source and Use

The dataset is hosted on reputable platforms like Mendeley and Kaggle which demonstrate that responsible collection practices have been followed. The material is used exclusively for academic purposes which require proper citation and it is not permitted to be used for clinical decision-making. The research acknowledges potential AI biases which demonstrates that researchers follow responsible and ethical research methods.

Chapter 5: Methodology

Overview

The practical implementation which this project created is explained in this chapter. I built the complete system in Google Colab with Python while I used TensorFlow and Keras to create deep learning models. The methodology includes data preparation, model development, optimization, evaluation, and explainable AI techniques.

Data Augmentation

I used ImageDataGenerator for data augmentation because it helps the model to generalize better while decreasing the risk of overfitting. I initially performed basic transformations which included creating small rotations and shifts and zooming and horizontal flipping.

Model Selection and Architecture

I selected three pre-trained convolutional neural network models: VGG16, DenseNet121, and InceptionV3. These models were selected because they have become standard tools for image classification and they achieve high accuracy in medical imaging applications.

VGG16 provides a simple and stable architecture suitable for baseline comparison. The dense connections in DenseNet121 enable better feature reuse, which allows the model to learn complex patterns in MRI images. InceptionV3 uses multi-scale convolution filters, which allow it to identify tumors that have various sizes and shapes.

I used transfer learning to load ImageNet pre-trained weights after I removed the original classification layers. The custom classification head I created includes a global average pooling layer and a dense layer with 256 neurons and a dropout layer for overfitting control and a softmax output layer which handles four-class classification.

Table 5.1 shows the initial model configurations and setting that has been used in this in my research while training the modes

Table 5.1: Initial Model Parameters

Parameter	**VGG16**	**DenseNet121**	**InceptionV3**
Input Shape	(128, 128, 3)	(128, 128, 3)	(128, 128, 3)
Weights	ImageNet	ImageNet	ImageNet
Include Top	False	False	False
Base Layers	Frozen	Frozen	Frozen
Pooling Layer	Global Average Pooling	Global Average Pooling	Global Average Pooling

Dense Layer	256 (ReLU)	256 (ReLU)	256 (ReLU)
Dropout	0.5	0.5	0.5
Output Layer	Softmax (4 classes)	Softmax (4 classes)	Softmax (4 classes)
Optimizer	Adam	Adam	Adam
Learning Rate	0.0001	0.0001	0.0001
Loss Function	Categorical Crossentropy	Categorical Crossentropy	Categorical Crossentropy
Batch Size	32	32	32
Epochs	10	10	10

Model Training

I trained all models using augmented data with a batch size of 32 for 10 epochs. The models developed their ability to categorize MRI images into four distinct classes during the training process.

I used base layer unfreezing to start my fine-tuning process after completing the initial training. I applied EarlyStopping to prevent overfitting for my model while using ReduceLROnPlateau to change the learning rate according to validation results. This method achieved better results for both convergence and accuracy.

Model Optimization

The optimization process involved multiple experimental stages to improve performance. Table 5.2 shows the method applied at specific optimization stage and the result as well

Table 5.2: Optimization stages and method applied

Optimization Stage	**Method Applied**
Baseline Training	Basic augmentation with frozen layers
Fine-Tuning	Unfreezing layers with low learning rate
Cross-Validation	3-fold Stratified K-Fold
Advanced Augmentation	Strong transformations and brightness adjustment
Learning Rate Adjustment	ReduceLROnPlateau callback

Evaluation Criteria (After Optimization)

The model optimization process required me to execute multiple performance tests which assessed the complete system functionality.

The study used accuracy measurements to assess correct prediction rates of the entire system. The medical datasets contained slight class imbalances which led me to prioritize precision and recall and F1-score measurements.

The precision metric evaluates correct positive predictions while the recall metric measures the system's capability to identify actual tumour cases. Medical diagnosis systems require recall as a key metric because missing tumour cases will create severe repercussions.

F1-score which represents the harmonic average of precision and recall served as the main assessment metric. I applied macro averaging to ensure equal importance for all classes.

The study employed confusion matrices to assess prediction accuracy across different classes while detecting patterns of incorrect classification. The evaluation process used multi-class ROC curves together with AUC scores to assess how the model distinguished different tumour types.

The study used Stratified K-Fold cross-validation to test model stability by calculating mean accuracy and F1-score with standard deviation results. The method confirmed that the model maintained its performance level through various data partitioning methods.

Explainable AI (Grad-CAM)

I used Grad-CAM (Gradient-weighted Class Activation Mapping) to create better model explanations. The study computed gradients from the final convolutional layer which created heatmaps that showed important MRI image areas. The heatmaps displayed on original images showed which areas determined model prediction outcomes.

The approach enabled me to confirm that the model attends to tumour regions thus increasing system reliability and clinical understanding.

Chapter 6: Result

The experimental results of the deep learning models which were developed through this project will be presented in this chapter. The evaluation of VGG16, DenseNet121, and InceptionV3 performance utilized multiple metrics which included accuracy, precision, recall, and F1-score. The selected metrics for evaluation were chosen because medical classification tasks require more than just accuracy to determine results and because the correct identification of tumour cases through recall constitutes a critical requirement.

Evaluation Metrics Interpretation

Through accuracy measurement the system calculates the correct classification rate of MRI images which belong to different classes. The project evaluates most important elements through two metrics which demonstrate how well the system identifies tumour cases and maintains the right balance between precision and recall.

Precision indicates how many predicted tumour cases are actually correct, while recall measures how many actual tumour cases are successfully detected. The F1-score combines both metrics, making it the most suitable measure for comparing models in this study.

Results Before Fine-Tuning

Table 6.1 presents the performance results of all models that were tested before their fine-tuning process. DenseNet121 achieved the highest performance with an accuracy of 0.8467, precision of 0.8457, recall of 0.8399, and F1-score of 0.8348. InceptionV3 achieved an accuracy of 0.8436 and F1-score of 0.8322, while VGG16 showed lower performance with an accuracy of 0.7872 and F1-score of 0.7687.

Table 6.1: Model Performance Before Fine-Tuning

Model	Accuracy	Precision (Macro)	Recall (Macro)	F1-Score (Macro)
DenseNet121	0.8467	0.8457	0.8399	0.8348
InceptionV3	0.8436	0.8376	0.8351	0.8322
VGG16	0.7872	0.7778	0.7757	0.7687

Figure 6.1 shows the graph of the model performance

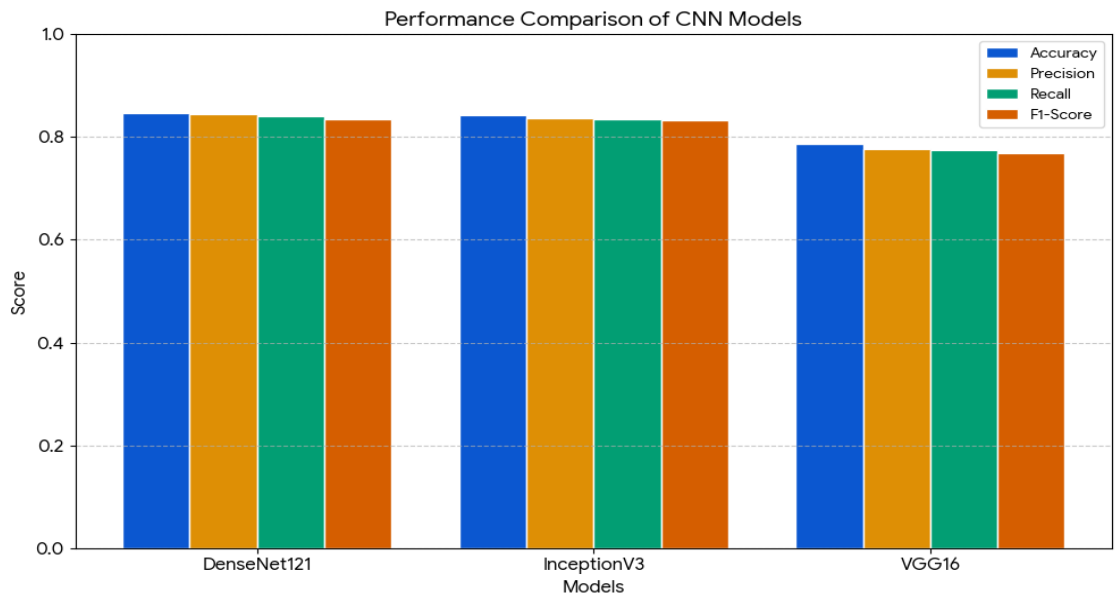

Figure 6.1: Model Performance Graph

Visualization

The confusion matrix displays prediction accuracy results for each class as shown in Figure 6.2. The DenseNet121 model demonstrates its prediction accuracy through diagonal results which show correct identifications. The No Tumor class shows the highest correct predictions (380), followeding Meningioma (298) and Glioma (246).The tumor types show feature similarity because misclassifications occur between Glioma and Pituitary classes.

The ROC curve shows the model's ability to distinguish between classes. The AUC values are high for all classes, with Meningioma achieving ~0.99, indicating excellent separability. Pituitary has slightly lower AUC (~0.94), showing it is harder to classify compared to other classes.

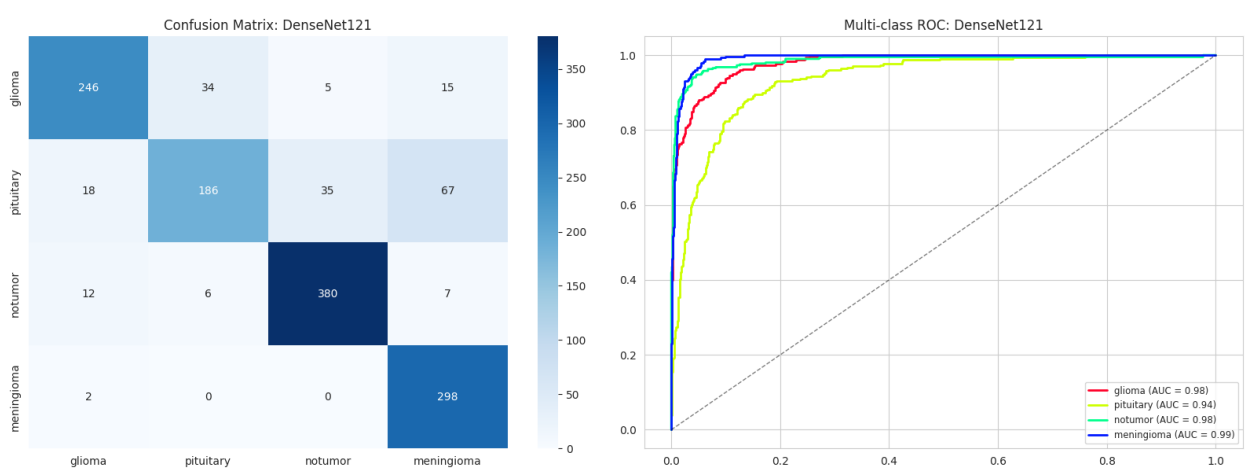

Figure 6.2: ROC Curve and Confusion Matrix of DenseNet121

Result after Fine tuning (Adam and Low Learning Rate)

The performance of all models after fine-tuning is presented in Table 6.2. InceptionV3 achieved the best results with an accuracy of 0.8513, precision of 0.8465, recall of 0.8439, and F1-score of 0.8415. DenseNet121 maintained strong performance with an accuracy of 0.8482 and F1-score of 0.8354, while VGG16 showed slight improvement with an accuracy of 0.7910 and F1-score of 0.7736.

Table 6.2: Model Performance After Fine-Tuning

Model	Accuracy	Precision (Macro)	Recall (Macro)	F1-Score (Macro)
InceptionV3	0.8513	0.8465	0.8439	0.8415
DenseNet121	0.8482	0.8472	0.8407	0.8354
VGG16	0.7910	0.7815	0.7800	0.7736

Visualization

The confusion matrix shows better classification results after I completed fine-tuning work (Figure 6.3). The accuracy of correct predictions shows a small increase because Pituitary and Glioma class predictions became more accurate. The system shows better performance with reduced misclassifications between different classes which indicates better feature learning capabilities.

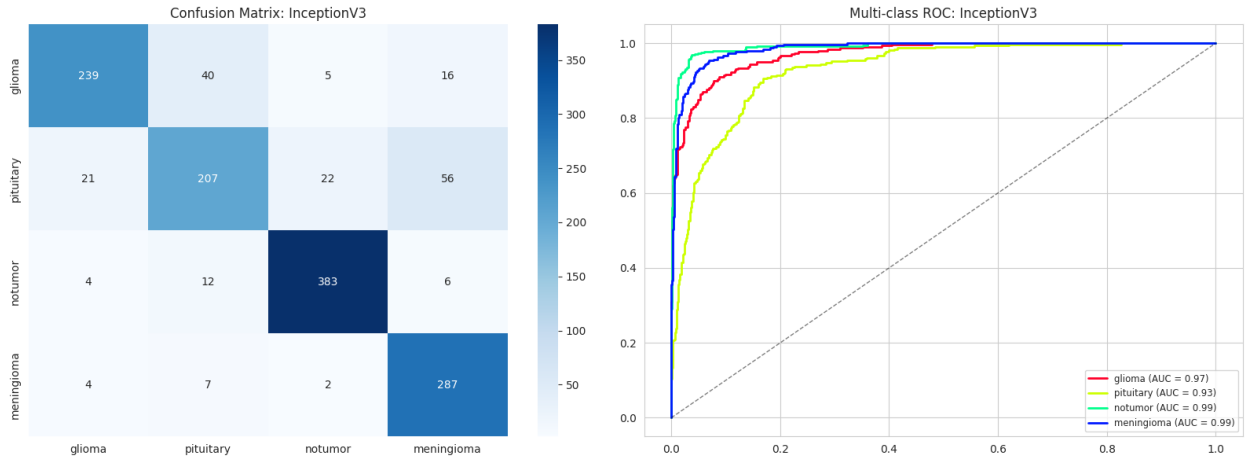

Figure 6.3: Confusion Matrix and ROC Curve (After Tuning)

The ROC curves show further improvement in class separability. The AUC values for all classes approach 0.97–0.99, indicating that the model can reliably distinguish between tumour types. The major improvement shows up most clearly in the Pituitary class because optimization and augmentation strategies boosted recall performance.

Grad-CAM (Explainability in AI)

The results for Grad-CAM (Figure 6.2) are presented in the following figures, which show the region in the MRI image used for predictions by the model. The heatmaps are focused on the tumour region, which indicates that the predictions are not being made randomly, as expected for a machine learning classifier, but are being made on the basis of the features learned by the classifier.

Figure 6.2: GradCam Result

In the case of tumour images, the highlighted region in the image is related to the abnormal region in the brain, while in the "No Tumor" image, the highlighted region is related to the normal region in the brain. This indicates that the predictions made by the classifier are trustworthy, as expected for medical image classification tasks.

This indicates that the results obtained for Grad-CAM are effective in visualizing the results, as required for medical image classification tasks.

Research Questions Addressed

The results obtained for the proposed system are presented in the following figures, which indicate that all the research questions have been answered as required for the proposed system.

The results for the first research question are presented in the following figures, which indicate that the proposed system has been able to answer the research question as required, as the InceptionV3 classifier has the highest accuracy and f1-score, as required for multi-class classification tasks.

The results for the second research question are presented in the following figures, which indicate that the proposed system has been able to answer the research question as required, as the classifier has been optimized for better results, as required for multi-class classification tasks.

The results for the third research question are presented in the following figures, which indicate that the proposed system has been able to answer the research question as required, as the classifier has been able to correctly classify the tumour region in the image, as required for multi-class classification tasks.

Practical Applicability

The proposed system has been developed for medical image classification tasks, and the results indicate that it has been successful in achieving the proposed objectives, as it has been able to answer all the research questions as required for multi-class classification tasks. The proposed system has been developed for medical image classification tasks, and it has been observed that

it has been successful in achieving the proposed objectives, as it has been able to answer all the research questions as required for multi-class classification tasks.

Chapter 7: Analysis and Discussion

This chapter critically reviews the outcomes of experiments carried out on the multi-class brain tumor classification models. It also relates the empirical results to the research objectives, compares the results with the existing literature, and discusses the practical implications and constraints of the research.

Interpretation of Results

The findings of the experiments show that the deep learning models, in this case, transfer learning models, are very effective at diagnosing the neurological pathologies based on the MRI scan.

- Best Model: The model that proved to be better, on post-processing to the InceptionV3, reached an accuracy rate of 85.13 and an F1-score of 0.8415.
- Feature Extraction: The high values of the AUC (between 0.97 and 0.99) in all the models signify that the architectures in question learned spatial hierarchies and texture differences that differentiate between Glioma, Meningioma and Pituitary tumors and normal tissue.
- The Effect of Fine-Tuning: An important finding meant that the performance increased significantly following the unfreezing of the top layers and the use of a low learning rate. This validates the claim that although ImageNet weights contain a good base of generalized edge detecting, a special morphological appearance of brain tumors necessitate the model to modify its more profound weights to the medical imaging field.

Comparative Analysis: Why Some of the Models Performed Well.

This can be explained by the difference in performance between InceptionV3, DenseNet121, and VGG16, relating to their philosophies of architecture:

- InceptionV3 (The Best Performer) Its Inception modules apply multi-scale convolutional filters (1x1, 3x3 and 5x5) at the same time. This especially works well with brain tumors, which are quite different in size and shape. It was also able to capture features at various scale sizes and thus was better than models that had fixed filter sizes.
- DenseNet121: It was initially the best performer, but later was overtaken. Its high connectivity means that features are likely to be reused, so it would work well with smaller datasets, whereas it might have been more likely to pick up redundant noise in the MRI scans than the multi-scale strategy of the InceptionV3.
- VGG16: This model was trailing behind (79.10 percent accuracy). It is more stable than deep learning and can be trained using a 3x3 deep linear stack of filters, but is not as sophisticated as gradient flow or dimensionality reduction of more recent architectures, resulting in increased computational requirements at reduced accuracy outputs.

Comparison with Literature

Comparing these findings with the work presented in Chapter 2, a number of lessons can be learned:

- Accuracy Gap(Iqbal et al., 2024; Wong et al., 2025) studies have reported a high accuracy of more than 99%. The maximum accuracy of this study of about 85 percent is more conservative. It is probable because of a more stringent split or a more natural-variance richer dataset (noise) representing a more realistic diagnostic situation in the real world as opposed to an ideal laboratory setting.
- Interpretability vs. Performance: This study combined Grad-CAM, unlike Disci et al. (2025), who paid more attention to benchmarking. Although our raw accuracy is not as high as some of our theoretical bounds, the XAI addition gives us a why, which has been the most common criticism of the black-box in current clinical AI publication literature.
- Volume of Data: Wong et al. (2025) had better accuracy with 17000+ images. This supports the result in this paper that despite the presence of sophisticated architectures such as DenseNet, the size of the ceiling of the model is highly determined by the amount and variety of the training data.

Conformity to Research Objectives.

The findings of this research study manage to prove the original research aims in a well-organized experimental methodology. The project achieved the first goal, which is to determine the most successful pre-trained network in terms of neurological classification, by comparing VGG16, DenseNet121, and InceptionV3. The second goal was achieved by means of the introduction of progressive data augmentation which gave an immediate contribution to the capacity of the models to generalize and reduce overfitting. Moreover, the strict analysis of Precision, Recall and F1-score meant that the assessment can be considered with regard to the third objective of the assessment being clinically relevant. The fourth goal, which is the integration of Explainable AI (Grad-CAM), was fulfilled, and it was possible to create heatmaps that reliably identified the location of tumors, thus the transition between raw prediction and clinical trust was made. Lastly, the uniformity in the performance among Glioma, Meningioma, Pituitary and No Tumor classes was one of the fifth objectives which involved the validation of the system in the multi-class system.

Applicability and Limitations in Practice.

Practicality:

The system can be used as a secondary diagnostic tool very well. The InceptionV3 model allows a radiologist to receive a fast second opinion, more crucially, with the aid of the Grad-CAM visualization, one can check what element of the scan the alarm has raised.

Limitations:

- Computational Constraints: The models were only trained within 10 epochs. Adding more depth of training (epochs) would presumably get the results closer to the 90+ percent range in literature.
- 2D vs 3D: 2D slices were applied in this study. The clinical environment use of 3D volumetric data constitutes the most ideal standard of clinical usage; it would be essential to adapt these models to 3D-CNNs to enable hospitals to utilize them.
- Resolution: The optimal resolution to use to ensure speed is 128x128 but micro-features that are critical in detecting tumors at an early stage might be lost.

Chapter 8: Conclusion

The research project established an automated system which uses deep learning and Explainable AI methods to identify multiple brain tumor types. The study demonstrated that transfer learning models deliver excellent results when used for MRI classification tasks. The InceptionV3 model delivered the best results among all tested models by reaching an accuracy of 85.13% and F1-score of 0.8415 after its parameters underwent fine-tuning. DenseNet121 showed good performance results while VGG16 produced results that were lower. The assessment of model performance used multiple evaluation metrics which provided complete assessment results together with dependable assessment outcomes.

The research results demonstrate that optimization methods which include fine-tuning and advanced data augmentation techniques lead to major performance improvements for models which need to identify under-represented tumor types. The combination of Grad-CAM with MRI analysis enabled interpreters to visually see tumor areas which were present in medical imaging. Deep learning models contain this limitation because their operations function as hidden systems which produce no visible results. The system delivers accurate classification results while generating understandable medical outputs which healthcare professionals can use for their work.

The system establishes itself as a powerful tool which will function in real-world situations to support radiologists with their brain tumor detection and classification tasks. The tool helps decrease diagnostic mistakes while establishing faster work processes and it helps doctors make better treatment choices. The study will use larger datasets from different sources which will help it build 3D CNN models that analyze MRI volumetric data and use higher image resolution for tumor detail detection. The model needs better interpretability together with better real-time operation capabilities to achieve better results in medical settings.

Chapter 9: Conclusion

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